

TASKS:

- 1. Install Nessus Essentials.**
- 2. Complete initial setup.**
- 3. Create a Basic Network Scan.**
- 4. Scan:**
 - **Localhost**
 - **Personal Test Machine**
 - **Lab Environment**

Observe:

- **Vulnerabilities Found**
- **Severity Levels**
- **CVSS Scores**
- **Affected Services**
- **Recommendations**

Assignment:

Create a report containing:

- 1. Critical Findings**
- 2. High Findings**
- 3. Medium Findings**
- 4. Low Findings**
- 5. Remediation Steps**
- 6. Screenshots**

Research the top 5 vulnerabilities detected and explain:

- **Impact**
- **Risk**
- **Fix**

Step 1: Download Nessus Essentials

Objective

Download the Nessus Essentials vulnerability scanner from the official Tenable website.

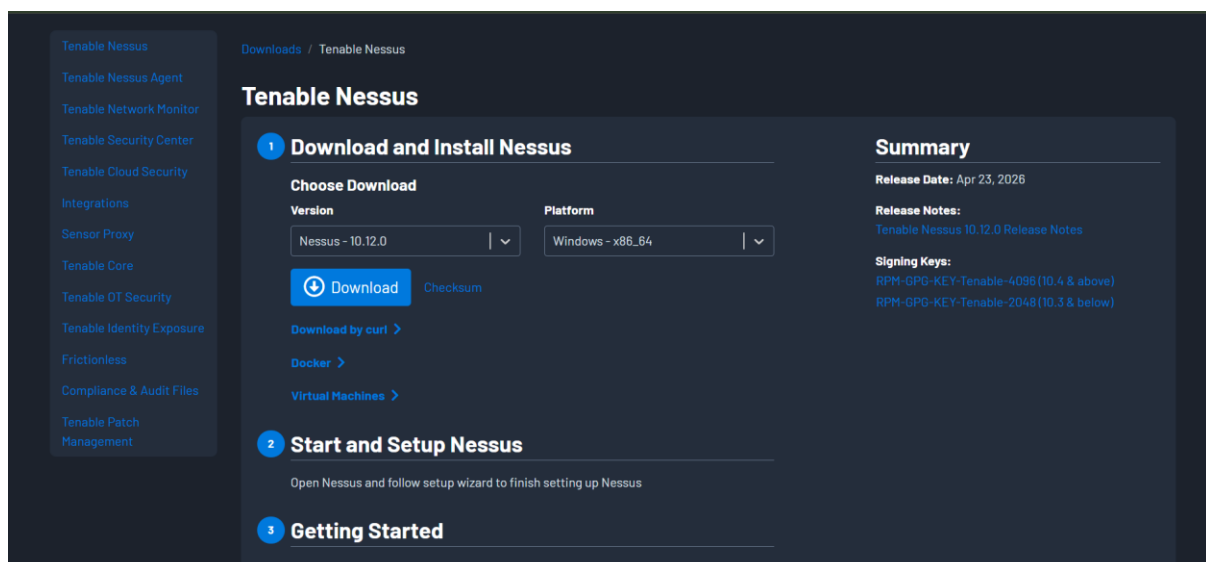
Procedure

1. Open a web browser (Chrome, Edge, or Firefox).
2. Visit the official Tenable Nessus Essentials page:
[Tenable Nessus Essentials Download Page](#)
3. Click "**Download Nessus Essentials**".
4. Fill in the registration form:
 - a. First Name
 - b. Last Name
 - c. Email Address
 - d. Country (if required)
5. Submit the form.
6. Check your email inbox for the Nessus Essentials activation code.
7. Download the installer suitable for your operating system:
 - a. Windows (.exe)
 - b. Linux (.deb or .rpm)
 - c. macOS (.dmg)

Outcome

- The Nessus Essentials installer is downloaded successfully.
- An activation code is received via email from Tenable.

Screenshot :



Step 2: Install Nessus Essentials

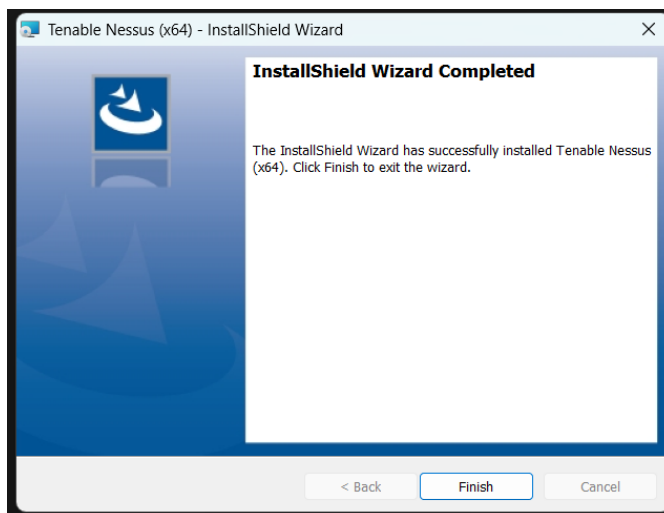
Procedure

1. Locate the downloaded Nessus installer file.
2. Double-click the installer to start the installation.
3. If a User Account Control (UAC) prompt appears, click **Yes**.
4. Follow the installation wizard:
 - a. Click **Next**
 - b. Accept the License Agreement
 - c. Choose the installation location
 - d. Click **Install**
5. Wait for the installation process to complete.
6. Click **Finish** after installation.

Expected Outcome

- Nessus Essentials is installed successfully.
- Nessus service starts automatically in the background.

Screenshot :



Step 3: Access Nessus Setup Page

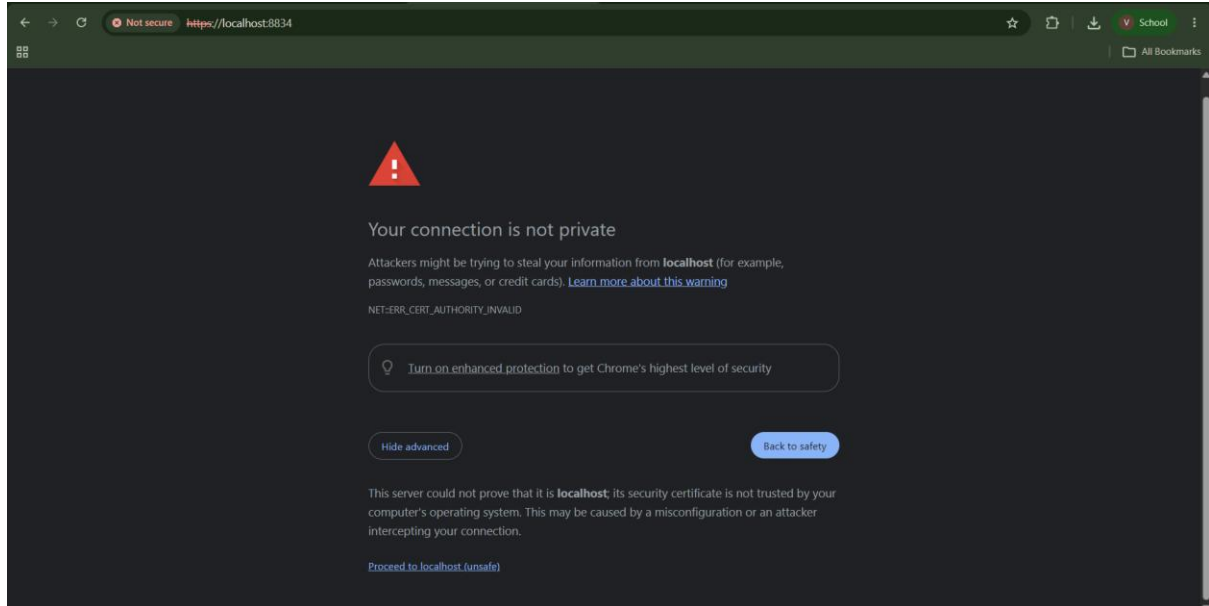
Procedure

1. Open a web browser (Chrome, Edge, or Firefox).
2. In the address bar, enter:
3. `https://localhost:8834`
4. Press **Enter**.
5. A security warning may appear because Nessus uses a self-signed certificate.
6. Click:
 - a. **Advanced**
 - b. **Proceed to localhost (unsafe)**
7. Wait for the Nessus welcome page to load.

Observation

The Nessus web interface was successfully accessed through `https://localhost:8834`. The browser displayed the Nessus Welcome page, indicating that the service was running correctly.

Screenshot:



Step 4: Select Nessus Essentials

After clicking "**Proceed to localhost (unsafe)**", the Nessus setup wizard should open.

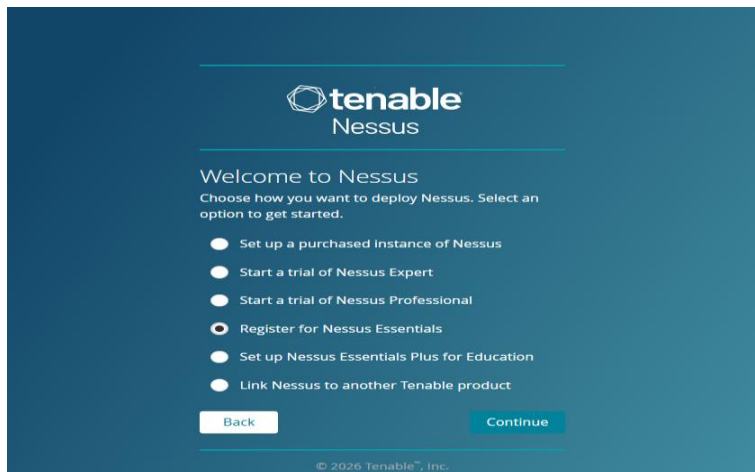
What to do

1. On the Welcome page, you will see Nessus product options.
2. Select:

Nessus Essentials

3. Click **Continue**.

Screenshot :



Step 5: Create Administrator Account

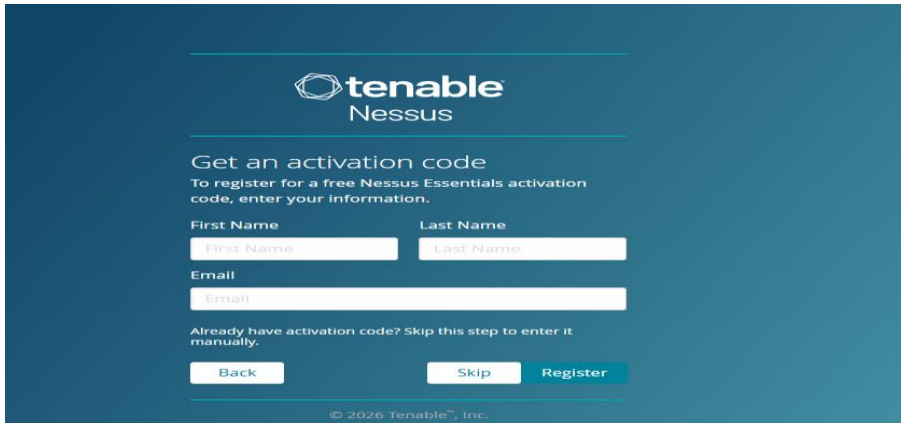
Objective

Create an administrator account to manage Nessus scans and settings.

Procedure

1. On the setup page, locate the **Create Administrator Account** section.
2. Click **Submit** or **Continue**.

Screenshot :



Step 6: Activate Nessus

Objective

Activate Nessus Essentials using the activation code received from Tenable.

Procedure

1. Open the activation page displayed after account creation.
2. Locate the **Activation Code** field.
3. Enter the activation code received in your email from Tenable.
4. Click **Continue** or **Activate**.
5. Wait for Nessus to verify the activation code online.

Step 7: Plugin Download and Initialization

Objective

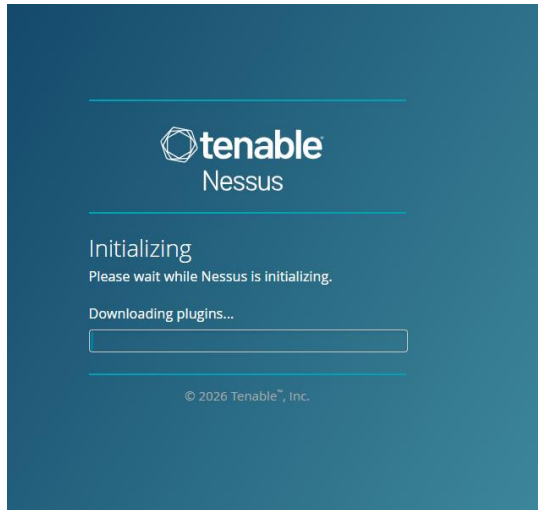
Download and initialize Nessus vulnerability plugins.

Procedure

1. After activation, Nessus automatically starts:

- Downloading plugins
 - Compiling plugins
 - Initializing the scanner
2. Wait for the process to complete.
3. This may take **10–20 minutes** depending on internet speed and system performance.

Screenshot:



Step 8: Access the Nessus Dashboard

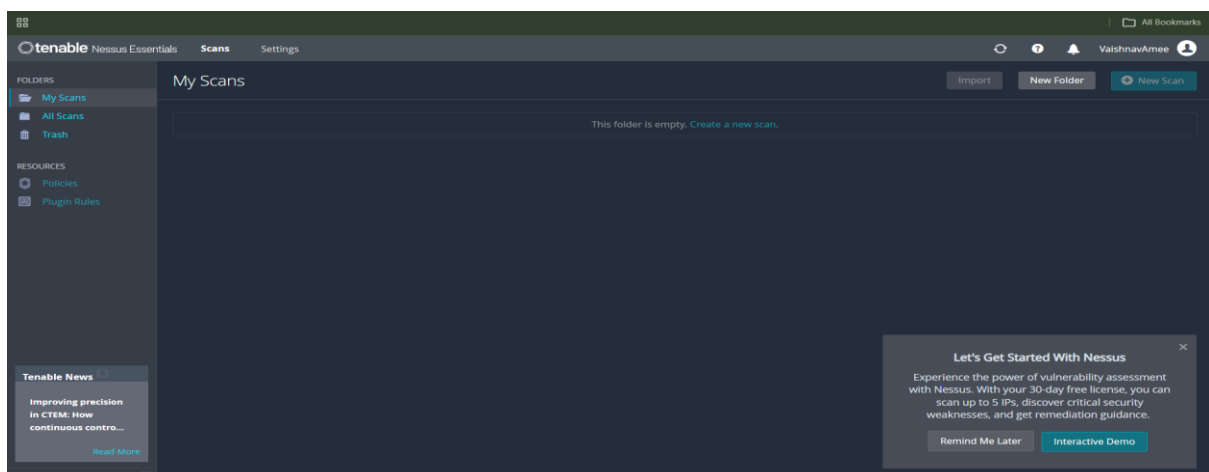
Objective

Verify that Nessus has been successfully installed, activated, and initialized.

Procedure

1. Wait for the plugin download and initialization process to complete.
2. Nessus will automatically redirect to the login page.
3. Log in using the administrator credentials created earlier.
4. After successful login, the **Nessus Dashboard** (Home Screen) will appear.

Screenshot :



Target 1- Create a Basic Network Scan

Objective

Create a new Basic Network Scan in Nessus to assess the security of a target system.

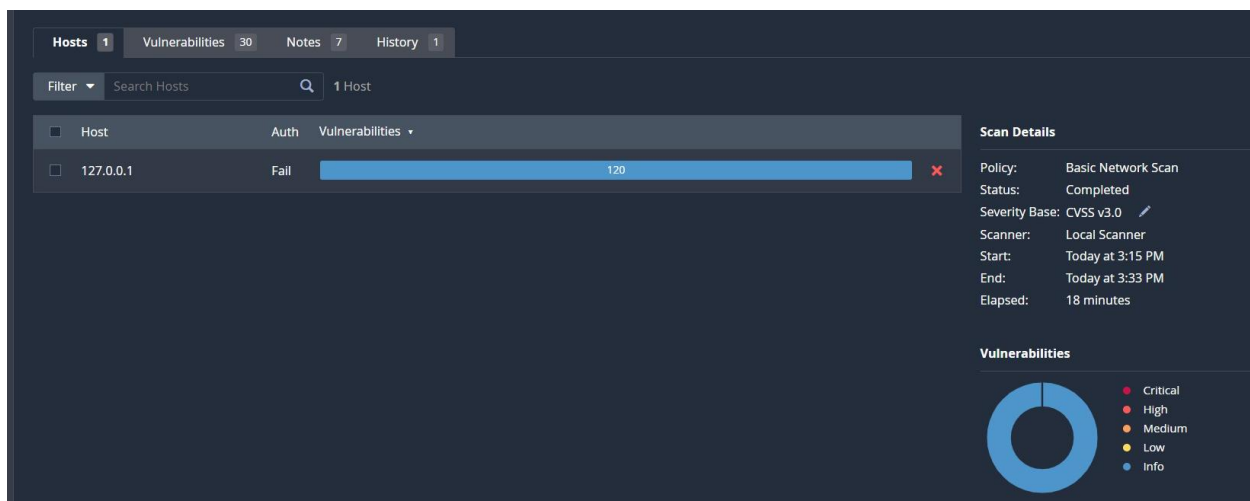
Procedure

1. Log in to the Nessus Dashboard.
2. Click the **New Scan** button (usually located in the upper-right corner).
3. A list of scan templates will appear.
4. Select:
Basic Network Scan
5. The scan configuration page will open.
6. Click **Save**.

Observation

A Basic Network Scan template was selected and configured to scan the localhost address (127.0.0.1). The scan configuration was saved successfully.

Screenshot :



Target 2 – Personal Test Machine (Windows)

IP Address:10.71.170.76

Purpose:

To identify active services, open ports, operating system information, and security vulnerabilities present on a Windows-based test machine connected to the network.

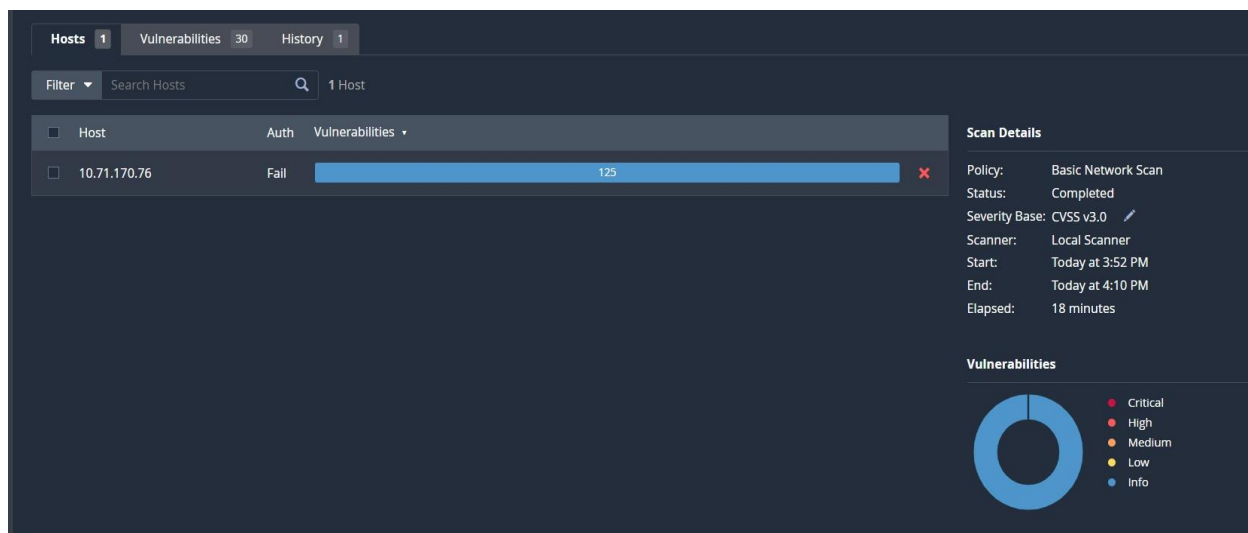
Procedure:

- Open Nessus Essentials.
- Create a Basic Network Scan.
- Enter the target IP address: **10.71.170.76**.
- Save the scan configuration.
- Launch the scan.
- Wait for the scan to complete.
- Review vulnerabilities, severity levels, CVSS scores, affected services, and remediation recommendations.

Observation:

The scan successfully identified running services, open ports, software versions, and potential security vulnerabilities on the Windows test machine. Vulnerabilities were categorized into Critical, High, Medium, Low, and Informational severity levels based on their CVSS scores. Nessus provided detailed descriptions and remediation recommendations for each identified issue.

Screenshot :



Target 3 – Lab Environment (Kali Linux Virtual Machine)

Target:

Kali Linux Virtual Machine

Purpose:

To assess the security posture of a Kali Linux virtual machine by identifying active services, open ports, potential vulnerabilities, and configuration weaknesses within the lab environment.

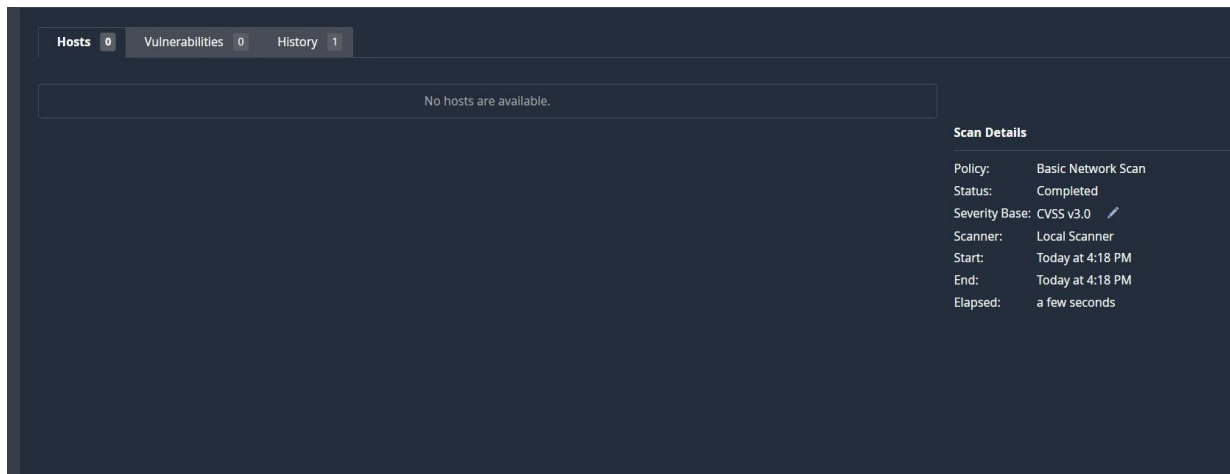
Procedure:

- Open Nessus Essentials.
- Create a new Basic Network Scan.
- Enter the IP address of the Kali Linux Virtual Machine.
- Save the scan configuration.
- Launch the scan.
- Wait for the scan to complete.
- Analyze the identified vulnerabilities and recommendations.

Observation:

The scan could not identify any reachable hosts because the Kali virtual machine was configured using VirtualBox NAT networking and was not reachable from the host system running Nessus.

Screenshot



Scan Results Summary

Severity Level	Findings
Critical	0
High	0
Medium	0
Low	0
Informational	30

Observation:

No Critical, High, Medium, or Low severity vulnerabilities were detected during the Nessus assessment. The scan identified informational findings related to service detection, network protocols, and system configuration details. These findings provide visibility into the target system's exposed services and network characteristics but do not represent immediate security risks.

Identified Services

The following services were detected during the assessment:

- HTTP Service
- SMB Service
- SSL Service
- TLS Service
- NetBIOS Service
- MySQL Service
- Windows Remote Services

These services were identified and analyzed to understand the system's network exposure, service configurations, and security posture.

- 1. Critical Findings:**No Critical severity vulnerabilities were detected during the assessment.
- 2. High Findings:**No High severity vulnerabilities were detected during the assessment.
- 3. Medium Findings:**No Medium severity vulnerabilities were detected during the assessment.
- 4. Low Findings:**No Low severity vulnerabilities were detected during the assessment.

Informational Findings

The following informational findings were detected:

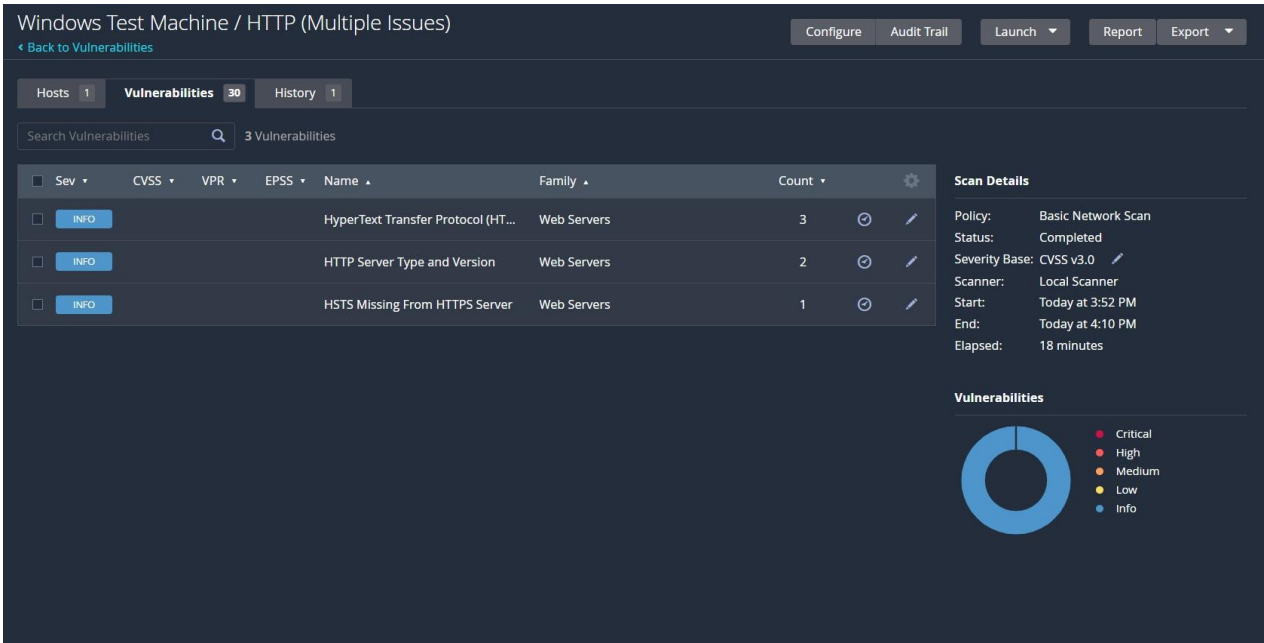
HTTP Findings

- HyperText Transfer Protocol (HTTP)
- HTTP Server Type and Version
- HSTS Missing From HTTPS Server

Observation:

These findings provide information about the web server configuration and the HTTP service running on the target system. The absence of HSTS indicates that browsers are not instructed to enforce HTTPS connections, but this was reported as an informational finding.

Screenshot :



SMB Findings

- Microsoft Windows SMB Service Detection
- Microsoft Windows SMB Native LAN Manager Detection
- Microsoft Windows SMB Version Detection
- Microsoft Windows SMB2 and SMB3 Detection
- Windows NetBIOS / SMB Remote Host Information Disclosure

Observation:

These findings identify the SMB services and protocols enabled on the target system. Information such as SMB versions and NetBIOS details was detected, helping administrators understand the system's network file-sharing configuration.

Screenshot :

The screenshot displays a web-based interface for a vulnerability scanner. The main title is "Windows Test Machine / SMB (Multiple Issues)". Below the title, there are tabs for "Hosts" (1), "Vulnerabilities" (30), and "History" (1). A search bar labeled "Search Vulnerabilities" shows "5 Vulnerabilities".

Sev	CVSS	VPR	EPSS	Name	Family	Count	
INFO				Microsoft Windows SMB Service ...	Windows	2	
INFO				Microsoft Windows SMB NativeL...	Windows	1	
INFO				Microsoft Windows SMB Version...	Windows	1	
INFO				Microsoft Windows SMB2 and S...	Windows	1	
INFO				Windows NetBIOS / SMB Remot...	Windows	1	

On the right side, there is a "Scan Details" section with the following information:

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Below the scan details is a "Vulnerabilities" section featuring a donut chart. The chart shows a distribution of vulnerability severity levels: Critical (red), High (orange), Medium (yellow), Low (green), and Info (blue). The "Info" category is the most prominent, represented by a large blue segment.

SSL Findings

- SSL Cipher Suites Supported
- SSL Perfect Forward Secrecy Cipher Suites
- SSL Cipher Block Chaining Cipher Suites

Observation:

These findings list the SSL cipher suites supported by the target system. The detected cipher suites can be reviewed to ensure that only strong encryption algorithms are enabled.

Screenshot :

The screenshot displays a security scanner interface for a 'Windows Test Machine / SSL (Multiple Issues)'. The interface includes a top navigation bar with buttons for 'Configure', 'Audit Trail', 'Launch', 'Report', and 'Export'. Below this, there are tabs for 'Hosts' (1), 'Vulnerabilities' (30), and 'History' (1). A search bar for 'Vulnerabilities' shows '3 Vulnerabilities'. The main table lists the following findings:

Sev	CVSS	VPR	EPSS	Name	Family	Count	
Info				SSL Cipher Suites Supported	General	2	
Info				SSL Perfect Forward Secrecy Cip...	General	2	
Info				SSL Cipher Block Chaining Cipse...	General	1	

On the right, the 'Scan Details' section provides the following information:

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Below the scan details, a 'Vulnerabilities' section features a donut chart showing the distribution of severity levels. The legend indicates the following counts:

- Critical: 0
- High: 0
- Medium: 0
- Low: 0
- Info: 3

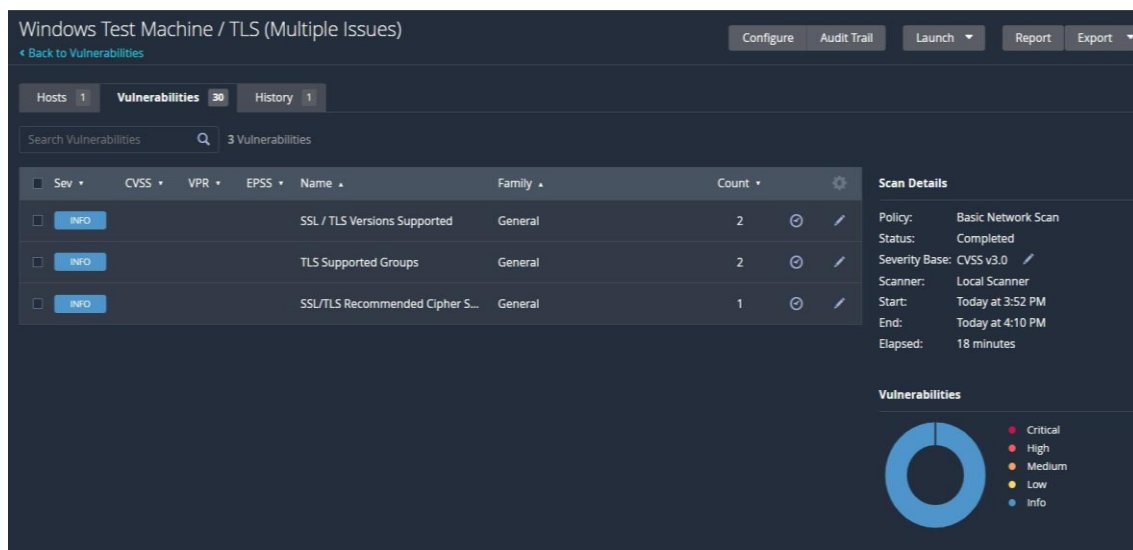
TLS Findings

- SSL/TLS Versions Supported
- TLS Supported Groups
- SSL/TLS Recommended Cipher Suites

Observation:

These findings provide details about the TLS protocols and cryptographic groups supported by the target system. The information helps evaluate the security of encrypted communications and verify compliance with recommended security practices.

Screenshot :



Top 5 Security Findings

1. HSTS Missing From HTTPS Server

Impact:

The website does not enforce the use of secure HTTPS connections for all visitors.

Risk:

Users may be vulnerable to protocol downgrade attacks, potentially allowing communication over unsecured HTTP connections.

Recommendation:

Configure HTTP Strict Transport Security (HSTS) to ensure browsers always connect using HTTPS.

2. HTTP Server Type and Version Disclosure

Impact:

Details about the web server software and version are publicly accessible.

Risk:

Attackers may use this information to identify known vulnerabilities associated with the detected software version.

Recommendation:

Suppress server version information and regularly update web server software to the latest supported release.

3. SMB Version Detection

Impact:

The scan identified the SMB protocol versions supported by the system.

Risk:

Legacy SMB protocols may introduce security concerns if they remain enabled.

Recommendation:

Disable outdated SMB versions and utilize SMBv2 or SMBv3 for secure file-sharing operations.

4. SSL Cipher Suites Supported

Impact:

The supported SSL/TLS encryption algorithms were successfully enumerated.

Risk:

Weak encryption ciphers, if enabled, may reduce the overall security of encrypted communications.

Recommendation:

Review enabled cipher suites and remove any outdated or weak encryption algorithms.

5. SSL/TLS Versions Supported

Impact:

Information regarding supported SSL/TLS protocol versions was disclosed.

Risk:

Older protocol versions may contain known security weaknesses and should be avoided.

Recommendation:

Configure services to support only modern protocols such as TLS 1.2 and TLS 1.3.

CVSS Information

The Common Vulnerability Scoring System (CVSS) is a standard framework used to evaluate the severity of security vulnerabilities.

Observations:

- No Critical severity vulnerabilities were identified.
- No High severity vulnerabilities were identified.
- No Medium severity vulnerabilities were identified.
- No Low severity vulnerabilities were identified.
- The assessment results consisted entirely of informational findings related to service identification and configuration details.

Remediation Recommendations

The following actions are recommended to further strengthen the security posture of the scanned systems:

1. Configure HSTS to enforce HTTPS communication.
2. Limit unnecessary disclosure of server information.
3. Remove support for SMBv1 where applicable.
4. Restrict SMB communication to SMBv2 and SMBv3.
5. Use only strong and secure SSL/TLS cipher suites.
6. Disable deprecated SSL/TLS protocol versions.
7. Apply security updates and patches on a regular basis.
8. Conduct periodic vulnerability assessments.
9. Disable services that are not required for operation.
10. Implement appropriate firewall and access-control policies.

Conclusion

The Nessus Essentials deployment and scanning activities were completed successfully. Security assessments were performed against localhost, a Windows-based test system, and a controlled laboratory environment. The scans revealed informational findings associated with web services, SMB configurations, SSL/TLS settings, and network service detection.

No Critical, High, Medium, or Low severity vulnerabilities were discovered during the assessment. Based on the scan results, the evaluated systems appear to be securely configured under the scope of the selected scan policy. Continued monitoring, regular patch management, and routine security assessments are recommended to maintain a strong security posture.

.